

World Bank Dhaka achieves 30.1% reduction in energy costs through HVAC and Solar Rooftop Optimization

CASE STUDY



BUILDING OVERVIEW

LOCATION

DHAKA, BANGLADESH

BUILT IN

2000s

Number of Floors

4

BUILDING CERTIFICATION LEVEL

GOLD

CONTROLLER TYPE

JOHNSON CONTROLS

EQUIPMENT CONTROLLED

• AIR HANDLING UNITS (AHUS) •
VARIABLE AIR VALVES (VAVS) •
OUTDOOR AIR & EXHAUST AIR FANS •
HOT WATER SYSTEM • CHILLED WATER
SYSTEM • SOLAR ROOFTOP

CONTEXT

A Modern Office Building in Southeast Asia

Situated in Bangladesh's Capital center, World Bank Dhaka's head office has been a great architectural construction. Faced with rising energy asset management costs and regulations, the building's facility operation team sought an innovative solution to comply with regulations, optimize energy asset management costs particularly in its chilled water loop, peripheral systems and rooftop solar. Client identified Enerzyz as the ideal solution.

PROCESS

Unified Management through Enerzyz OS

Enerzyz worked closely with the building's controls contractor and internal team to integrate Enerzyz OS with the existing Building Management System (BMS). This allowed the facility team to use their existing tool set and systems to observe the AI in use throughout their building.

Enerzyz OS was ingesting data through our universal control module, while communicating via BACnet IP for BMS and API for inverters. The roll-out was executed floor by floor, in close cooperation with the building's Project Manager, ensuring that WBG personnels remained comfortable and disruption-free throughout. This collaborative effort was key to the customer's positive experience and the project's long-term success.

Main Office Space and lobbies at the building require chilled water at all times for temperature and humidity control. Enerzyz used the design criteria in the centrifugal pump schedule to develop a system curve and determined the minimum flow rate to ensure just enough temperature and humidity for comfort. This data was used as another input for our AI to consider, continuously optimizing set temperature and valve positions while satisfying the required thermal demand of the micro climates room wise. In addition to that, String output anomalies and soiling insight on solar rooftop was provided for prompt action and cleaning needs.

RESULTS

Results for all stakeholders

Over a 6-month period in 2025, World Bank Dhaka realized substantial electricity, emissions, and cost savings. Our Optimization solution drove a 30.1% reduction in energy costs in HVAC-related electricity consumption along with increased solar throughput, saving \$19,124, and mitigating 99.62 tCO₂eq. These remarkable results were achieved seamlessly, with the facility engineers noting no disruptions to daily operations – another key objective of the project.

With a firm focus on occupant comfort and smooth office operations (two key priorities of the client), our team and autonomous AI solution delivered outstanding results. through collaboration and effective communication with the customer and their controls contractor, Enerzyz consistently provided value throughout the ongoing partnership. As a result, World Bank Dhaka not only realized significant savings, but also met regulatory demands and enhanced operational efficiency, driving value for the organizational goal. Point to note that, total manhour reduction in facility management has not been taken into account for this case study.

Discover more of our solution, Book a demo [here](#).



-\$19,124

Total cost savings in 6 months



-25.1%

Reduction in HVAC related electricity consumption



-156,387 kWh

Reduction in HVAC related electricity consumption



-99.62 tCO₂eq

Reduction in HVAC related emissions



+27.1%

Increase in Solar Rooftop Performance